

Iodine-125 Irradiation of Choroidal Melanoma

Clinical Experience

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Abstract: Twenty-nine patients with choroidal melanomas were treated with iodine-125 seeds in gold scleral plaques. Iodine-125 emits low energy 25 KeV photons which are attenuated by the side and back of the plaques so that the radiation is "directed" towards the choroidal tumor. The anticipated therapeutic benefits of using a low energy isotope and directional applicators have been realized. There has been a marked reduction in ocular complications in the nine medium and 20 large melanomas so treated. Our complication rate was 34% with a mean follow-up of 38 months. This is a lower complication rate, with longer follow-up, and in larger tumors than that reported with the use of cobalt-60, ruthenium-106 or ion beam. Three patients with large tumors and two patients with medium tumors died of metastasis. This compares favorably to results of similarly sized melanomas treated by enucleation. [Key words: brachytherapy, choroidal melanoma, eye melanoma, iodine-125, radiotherapy.] *Ophthalmology* 91:1700-1708, 1984

Iodine-125 is a low energy gamma emitting radioisotope. It has been used extensively in the radiation therapy of cancers of the prostate, lung and other sites.¹ The physical properties of a low energy isotope offer the possibility of shielding the radiation by incorporating these sources of radiation into scleral plaques made of high atomic number metals, such as gold. The plaque surrounds the Iodine-125 seeds except for the side that faces the sclera. Thus radiation is directed into the eye at the site of the choroidal melanoma, while the shielding protects normal ocular structures adjacent to the tumor.

This combination of a low energy isotope, Iodine-125, and a metal gold, plaque was tested in a rabbit

model.² The results were encouraging and led to this clinical trial. Since our goal was to show beneficial results in a reasonable period we chose to treat only medium and large size tumors. If we were to treat small tumors it would not be possible to show significant comparisons to survival rates from enucleation for 15 or 20 years. However, the mortality rates of medium and large choroidal melanoma are such that the comparisons should be reasonable at 5 years. The same argument holds for complications, that is, eyes with small tumors would be expected to suffer fewer radiation complications. Therefore, if one wants to compare techniques it is difficult if one compares results of only treatment of small tumors.

It is important to stress that this paper examines primarily the effects of iodine-125 as a form of local therapy. However, we do not feel that malignant melanoma of the choroid is a local disease and have therefore used adjuvant therapy in selected cases. The goal was to find a form of therapy that effectively sterilizes a choroidal melanoma, has the least number of complications, is cost effective and could be used by ophthalmologists in collaboration with radiation therapists at most hospitals.

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Presented at the Eighty-eighth Annual Meeting of the American Academy of Ophthalmology, Chicago, Illinois, October 30-November 3, 1983.

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